

Yael Prigat Goldfriend

Novel Food and Pharma Researcher. Driven by Sustainability and Innovation

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PROFESSIONAL PROFILE

Food researcher and R&D strategist with an MSc in Chemistry and a BSc in Biochemistry and Food Sciences, bringing extensive applied research experience in alternative proteins, novel ingredients, food structure design, sensory-material relationships, and analytical characterization of complex food matrices. Experienced in designing, leading, and translating research programs across industry-academic interfaces, including supervision of scientific teams, collaboration with academic researchers, development of internal knowledge resources, peer-reviewed publications, and patent-pending technologies. Motivated to contribute to mission-oriented research initiatives that advance resilient, scalable, sustainable, and health-oriented future food systems.

EXPERIENCE

Food Science Research Lead - Innovation & Technology | Redefine Meat

2021-2026

- Led long-term applied research on alternative meat systems, with emphasis on structure-function relationships, lipid/fat mimetic design, texture formation, and sensory performance.
- Designed and supervised experimental studies comparing animal-derived beef fat tissue characteristics with alternative fat systems to inform the development of meat analogue matrices.
- Directed a team of four food scientists, providing research supervision, methodological guidance, experimental planning, and interpretation of analytical and sensory data.
- Investigated novel ingredients for their effects on texture, flavor release, structural stability, and formulation functionality in complex food matrices.
- Conducted analytical characterization of beef cuts and meat analogue prototypes using mechanical, sensory, and physicochemical methods; translated findings into internal R&D knowledge resources and conference presentations.
- Collaborated with academic researchers on plant protein functionality and characterization, contributing to a peer-reviewed publication and patented technologies.
- Contributed to patent-pending research on fat tissue mimetics for alternative meat applications, with emphasis on sensory-material relationships and food structure design.

Food Research and Development Scientist | Zero Egg

2019-2021

- Led formulation research for plant-based egg alternatives, focusing on protein functionality, texture, stability, sensory performance, and application-specific behavior.
- Designed and implemented quality and functionality assays to evaluate shelf life, performance consistency, and product stability across storage and processing conditions.
- Investigated ingredient systems including proteins, stabilizers, thickeners, and functional additives to improve texture, thermal behavior, and performance in food applications.
- Evaluated pasteurization technologies for shelf-life extension, integrating microbiological, chemical, and functionality-based criteria.
- Developed allergen-free, high-protein, and clean-label formulations in response to nutritional, regulatory, and consumer constraints.
- Coordinated microbiological and chemical testing to support formulation decisions, quality control, and product validation.

EDUCATION

MSc, Chemistry | The Hebrew University of Jerusalem

2016-2018

- Thesis: "Extraction and Solubilization of Gossypol from Cotton Seeds by NSSL." Advisor: Prof. Nissim Garti. Thesis grade: 88.5
- Planning and executing a microemulsion formulation tailored for the research needs.
- Testing the system's stability using various quality and quantity methods.
- Developed expertise in colloidal systems, formulation chemistry, physicochemical characterization, and experimental design.

BSc, Biochemistry and Food Sciences | The Hebrew University of Jerusalem

2013-2016

- Interdisciplinary training in biochemistry, food science, chemistry, and biological systems.
- Foundation in food composition, food chemistry, analytical methods, and biochemical mechanisms relevant to food functionality and stability.

PEER-REVIEWED PUBLICATIONS

1. Novel plant proteins used in 3D printed meat analogues: relationship between protein physicochemical and functional characteristics. *European Food Research and Technology*. <https://doi.org/10.1007/s00217-023-04297-8>
2. Micro-characterization of modified microemulsions loaded with gossypol, pure and extracted from cottonseed. *Colloids and Surfaces B: Biointerfaces*. <https://doi.org/10.1016/j.colsurfb.2019.05.004>

SELECTED RESEARCH PROJECTS

- **Plant protein combinations for improved functionality:** mentored the research of the synergistic effect of combination of proteins on their functionality. IP: US20250338872A1
- **Fat tissue mimetics for meat analogue applications:** Studied animal fat tissue as a reference system for designing alternative fat structures with improved sensory and mechanical behavior. IP: US20260083146A1
- **Hydrocolloid films for different uses:** Creation of a extensive catalog of hydrocolloid films and their mechanical properties, to be used as a basis for connective tissue in meat analogues. IP: WO2023031915A1
- **Analytical characterization of beef cuts and meat analogue matrices:** Used mechanical, sensory, and physicochemical data to build internal research resources for food structure and texture design.
- **Shelf-life and pasteurization evaluation in plant-based egg alternatives:** Developed testing approaches to evaluate stability, functionality, and microbial/chemical quality under different processing conditions.

TECHNICAL AND ANALYTICAL METHODS

Analytical characterization	Texture Profile Analysis (TPA); rheology; GC-MS; spectrophotometry; colorimetry; Dynamic Light Scattering (DLS); Differential Scanning Calorimetry (DSC), FTIR
Food science methods	Ingredient functionality screening; shelf-life testing; QA/QC for formulation
Sensory and data interpretation	Sensory panel design; questionnaire preparation; test execution; results analysis; sensory-material correlations
Food systems expertise	Alternative proteins; food formulations; protein functionality; food colloids and emulsions; stabilizers, thickeners, proteins, colorants, and functional ingredients
Research leadership	Experimental planning; research supervision; mentoring; academic-industry collaboration; scientific writing and dissemination.

PATENTS AND INTELLECTUAL PROPERTY

- **WO2023031915A1- An edible hydrogel, method of production and uses thereof**
inventors: Shmuel ITTAH, Jonathan HAUSNER, Yael PRIGAT GOLDFRIEND, Moran Cohen, Daniel Dikovsky, Nissim DAVID, Eshchar BEN-SHITRIT
- **US20250338872A1 - Plant protein combinations and uses thereof**
Inventors: Darya ISRAELI, Gili ABRAMOV, Ofir BENJAMIN, Yael PRIGAT GOLDFRIEND
- **US20260083146A1- Edible lipid element, method of preparation and uses thereof**
Inventors: Yael PRIGAT GOLDFRIEND, Moran Cohen, Daniel Dikovsky

MENTORING, SUPERVISION, AND SCIENTIFIC COMMUNICATION

- Supervised and mentored four food scientists in experimental planning, analytical workflows, data interpretation, and research documentation.
- Trained interns and junior team members in sensory testing, texture analysis, ingredient screening, and formulation evaluation.
- Communicated analytical findings to interdisciplinary R&D teams and academic collaborators.

LANGUAGES

English: fluent/professional	Hebrew: native/mother tongue	Spanish – intermediate	German – beginner
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ADDITIONAL INFORMATION

EU citizen